Design, Implementation and Evaluation of a Relational Database and Business Intelligence Prototype for Consumer Price Inflation Analysis in Ghana

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Declaration

I declare that this dissertation is my own original work and has not been submitted for any other degree or other professional qualifications. All sources of information used in this research have been duly and properly acknowledged and referenced in accordance with academic standards. I can confirm that I have complied with the University's rules and regulations regarding academic integrity and the appropriate use of generative artificial intelligence tools.

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Date: 23rd April, 2026

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I wish to acknowledge the Ghana Statistical Service for maintaining and publicly releasing the CPI datasets that form the empirical foundation of this study. Without their commitment to open data, this research would not have been possible.

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Abstract

Consumer price inflation is a critical macroeconomic indicator that shapes economic decision-making across governments, central banks and businesses. In Ghana, monthly Consumer Price Index (CPI) data published by the Ghana Statistical Service (GSS) is widely used in institutional analysis yet typically processed through error-prone, manually intensive spreadsheet workflows. Existing public inflation dashboards prioritise visualisation over transparent data architecture, limiting the verifiability and reproducibility of outputs.

Hence, this study addresses that gap by designing, implementing and evaluating a prototype relational database and business intelligence (BI) system for CPI analysis, using Ghana as a representative developing country case study. A PostgreSQL v15 database in Third Normal Form was constructed from four StatsBank datasets covering January 2015 to December 2024, comprising 2,160 rows across four tables. An ETL pipeline, five SQL analytics views, and four interactive Power BI dashboards were built and evaluated through data integrity checks, query performance benchmarking, spreadsheet comparison and scenario-based testing. SQL queries returned results in under 600 milliseconds and analytical tasks requiring up to four minutes in Excel were completed in under 200 milliseconds.

The prototype demonstrates that a structured, transparent and low-cost alternative to spreadsheet dependency is technically feasible in resource-constrained institutional settings, and that the same design is transferable to other statistical domains and developing countries' contexts.

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List of Abbreviations

3NF	Third Normal Form
BI	Business Intelligence
CPI	Consumer Price Index
COICOP	Classification of Individual Consumption According to Purpose
CSV	Comma-Separated Values
DSR	Design Science Research
ETL	Extract, Transform, Load
GSS	Ghana Statistical Service
IMF	International Monetary Fund
MoM	Month-on-Month
SQL	Structured Query Language
YoY	Year-on-Year

Chapter 1

Introduction

1.1 Background

Consumer Price Inflation is among the most consequential economic indicators any government monitors. It shapes decisions about interest rates, wages, social support, and investment policy. In developing countries, where economies shift quickly and public institutions operate with constrained resources, the capacity to monitor inflation accurately is foundational to informed governance.

In Ghana, the Ghana Statistical Service (GSS) publishes monthly Consumer Price Index (CPI) data covering national headline inflation, category-level breakdowns across fifteen product divisions, and year-on-year and month-on-month rates of change. These publications are released as PDF bulletins and other downloadable file formats like Excel and JSON through the GSS website and its StatsBank platform respectively. While the data itself is consistent and well-maintained, the format creates a structural problem, as institutions that work with this data routinely do so through spreadsheets, copying figures manually and repeating the same process each month. That approach is time-consuming, error-prone, and provides no built-in mechanism for catching mistakes [1].

A second problem receives less attention. Platforms such as the IMF DataMapper and the World Bank Macro Poverty Outlook produce professional visualisations but prioritise the presentation layer over transparent documentation of the underlying data architecture, making independent verification difficult for any policymaker who needs to trust the figures they are using [4].

This study addresses both problems. It builds a prototype system that is structured, transparent, and practically achievable, one that documents every stage of the data journey from source to dashboard and evaluates its performance against the spreadsheet workflows it is designed to replace.

1.2 Problem Statement

The problem this study tackles has two connected dimensions. The first is the persistent dependence on spreadsheets for CPI analysis in institutional settings. Spreadsheets are accessible and require minimal infrastructure, which explains their continued use. But their limitations in handling large, evolving time-series datasets are well established. Monthly CPI data accumulates over years, categories shift, and meaningful analysis requires comparisons across both time and indicators. In that environment, spreadsheets are vulnerable as formulas get misapplied, versions multiply, and errors go unnoticed [8]. Every manual step in the process, copying a figure from a bulletin, constructing a formula, building a chart, is an opportunity for error with no structural means of detection.

The second dimension is architectural opacity. Existing public inflation dashboards focus on visual output rather than on the ingestion, transformation, and governance of the data that produces those outputs. This makes it difficult to verify figures or reproduce the analysis independently. Trust in the results is therefore either undermined by the risk of human error, or uncertain because the underlying process cannot be examined [9].

Together these two problems define the gap this study addresses: CPI analysis in developing country institutional contexts lacks infrastructure that is both reliable and transparent.

1.3 Research Aim

The overall aim of this study is to design, implement and evaluate a prototype relational database and business intelligence system for analysing Consumer Price Inflation data in Ghana.

1.4 Objectives

To achieve this aim, the study pursues the following structured objectives:

- Review and analyse the structure, format, and consistency of monthly CPI publications released by the GSS between 2015 and 2024.
- Define functional and non-functional system requirements for a CPI analysis platform suited to institutional users.
- Design a relational database schema in Third Normal Form (3NF) to structure CPI data for efficient storage, validation, and querying.
- Implement the schema in PostgreSQL v15, configuring tables, primary and

foreign keys, constraints, and indexes suited to time-series analysis.

- Build a structured ETL pipeline using SQL scripts and Power BI Desktop to ingest and transform CPI data.
- Develop four interactive Power BI dashboards covering headline CPI trends, category-level indices, year-on-year and month-on-month changes, and multi-period trend exploration.
- Benchmark system performance by comparing query execution times and report generation speed against equivalent spreadsheet workflows.
- Evaluate system quality through query performance tests, reconciliation with official GSS figures, and structured scenario-based dashboard testing.
- Reflect critically on design and implementation decisions, noting limitations, trade-offs, and directions for future development.

1.5 Justification

This study is motivated by a practical and well-documented gap. Developing country public institutions manage significant volumes of statistical data, CPI included, but typically lack the infrastructure to handle that data in a structured and auditable way [2]. The result is a pattern in which analysis depends either on error-prone manual workflows, or on external platforms whose internal workings are not visible or verifiable [4].

CPI data makes a particularly strong case for this kind of intervention. It is published regularly, accumulates over time, spans multiple categories, and feeds directly into high-stakes decisions, making it exactly the kind of dataset that benefits from a properly structured database and suffers most when the tools used to analyse it are inadequate.

Ghana provides a concrete, accessible, and representative setting. The GSS publishes monthly CPI data consistently and makes it freely available, providing a complete and reliable dataset spanning decades. The institutional conditions in Ghana, including public sector spreadsheet dependency, constrained technical infrastructure, and a growing need for data-driven governance, reflect patterns seen across developing country contexts more broadly [4]. The system designed in this study is therefore not intended solely for Ghana, but offers a transferable model relevant to similar institutional environments elsewhere.

1.6 Methodology Overview

This study uses a Design Science Research (DSR) approach. DSR is a well-established methodology in information systems research that focuses on building and evaluating practical solutions to real problems rather than simply describing or explaining them [12]. The artefact produced in this study is a working prototype consisting of a relational database, a documented ETL pipeline, and four interactive BI dashboards. The system is evaluated through query performance benchmarking, reconciliation with official GSS data, and scenario-based dashboard testing. A full account of the methodology and the decisions behind it is provided in Chapter 3.

1.7 Dissertation Structure

The remainder of this dissertation proceeds as follows. Chapter 2 reviews the relevant literature across six areas: CPI data and inflation monitoring, spreadsheet risk, relational databases and data governance, business intelligence, digital transformation in developing country contexts, and design science research. It closes by identifying the research gap this study addresses. Chapter 3 presents the research design, covering the DSR methodology, data sources, system requirements, tool selection, and evaluation strategy. Chapter 4 documents the full system implementation. Chapter 5 presents the testing process and its results. Chapter 6 discusses the findings in relation to the problem, the literature, and the system's strengths and limitations. Chapter 7 draws conclusions and identifies directions for future development. Chapter 8 offers a critical self-evaluation of the project process and outcomes.

Chapter 2

Literature Review

2.1 CPI Data and Inflation Monitoring

Consumer Price Inflation is measured through the Consumer Price Index, a statistical tool that tracks price changes across a defined basket of goods and services over time. In Ghana, the GSS publishes monthly CPI figures covering national headline inflation and category-level breakdowns across fifteen COICOP divisions. These publications are made available as PDF bulletins and Excel files through the GSS website and StatsBank platform. While this represents a consistent and publicly accessible data release, the formats themselves present a foundational limitation: they are designed for dissemination, not for systematic analysis. Each bulletin is a standalone document, meaning that comparing trends across months or across categories requires opening multiple files simultaneously and constructing comparisons manually, a process that is both time-consuming and increasingly error-prone as the volume of data grows [5].

2.2 Spreadsheet-Based Analysis and Its Risks

The dominant institutional response to this limitation has been the spreadsheet. Spreadsheets are familiar, require no specialist infrastructure, and can be adopted without formal technical training. These qualities explain their continued use in public institutions across developing country contexts [2]. Their limitations in handling large, evolving datasets are, however, well established. Panko [1] demonstrated that even relatively simple spreadsheet models carry a surprisingly high rate of undetected errors, challenging the common assumption that a spreadsheet output is correct unless something visibly goes wrong. Powell et al. [7] extended this argument, showing that errors in operational spreadsheets can have significant downstream consequences for decision-making, particularly when outputs are used directly in planning and policy contexts. Grossman and Özlük [8] further highlighted that without formal engineering practices such as version control, structured validation, and documented logic, spreadsheet-based workflows remain fundamentally difficult to audit and reproduce.

What makes this problem particularly relevant for CPI analysis is the nature of the data itself. Monthly figures accumulate over years, categories evolve, and in Ghana's case the introduction of new base years at different points in time means that figures from different publication cycles require careful contextualisation before comparison. The cognitive reality compounds the technical one: the human mind tends to perceive data as correct unless it is actively searching for a specific discrepancy [1]. A misplaced decimal, an entry in the wrong cell, a skipped row, these are easy to produce and difficult to find, especially when the person reviewing the work is the same person who produced it.

2.3 Relational Databases and Data Governance

The established response in information systems research is the relational database. By enforcing a structured schema, applying constraints at the point of data entry, and separating raw data from analytical computation, relational databases address the core weaknesses of spreadsheet workflows [3]. Abraham et al. [3] argue that formalised data governance, of which structured database design is a foundational component, is essential for institutional data maturity. Kitchin [6] takes a broader view, arguing that datasets must be understood not as neutral objects but as components of information systems that require governance, lifecycle management, and documented provenance. Together, these perspectives establish that the problem with spreadsheet-based CPI analysis is not simply one of tool choice. It is a governance problem, and it requires a governance response.

2.4 Business Intelligence and Dashboarding

Business intelligence tools represent the analytical and presentational layer of that response. Chen et al. [10] established that BI tools generate their greatest value when backed by mature data architectures rather than used as standalone visualisation instruments. Kimball and Ross [11] provided the foundational framework for dimensional data modelling, particularly the star schema approach, which enables efficient querying and consistent reporting across large datasets. Shollo and Galliers [9] more recently argued that the gap between data and actionable insight is not closed by dashboards alone, but by the quality and transparency of the systems that feed them.

A growing body of applied BI research has demonstrated the value of these tools across public sector and data-intensive settings. Adeshina [15] showed that federated analytical architectures can support complex institutional data needs in public health surveillance. Gurcan et al. [16] confirmed the field's shift towards integrated, architecture-aware approaches over two decades. Petersen et al. [17] found measurable reductions in reporting time following BI adoption in a municipal context. Rabiei et al. [18] and Sanabia-Lizarraga et al. [19] further demonstrated the tool's value in health surveillance and agricultural trade analysis respectively. Collectively, these works confirm the analytical potential of BI in institutional settings while sharing a common limitation. They focus on outputs without fully documenting the underlying data architecture, ETL processes, or schema design decisions that make those outputs verifiable.

2.5 Digital Transformation in Developing Country Contexts

This architectural gap is further widened by the institutional context in which CPI analysis takes place. Nielsen and Sahay [2] documented how developing country governments face structural constraints in building sustainable digital data infrastructure, including limited technical capacity, fragmented systems, and a tendency to adopt surface-level digital tools without reforming the governance structures beneath them. Mergel et al. [4] found that digital transformation in public institutions often prioritises visible, presentation-focused outputs over back-end architectural reform. This pattern explains why polished dashboards can coexist with poorly governed data pipelines, and why the analytical value of BI tools is so frequently constrained by the quality of the data systems behind them.

2.6 Design Science Research

The Design Science Research methodology provides the appropriate framework for responding to this kind of problem. Rather than analysing or theorising about the gap, DSR calls for the creation of a practical artefact that is then rigorously evaluated in context [12]. Hevner et al. [12] established DSR as a legitimate and rigorous paradigm in information systems research, emphasising that the artefact must address a genuine problem and be evaluated against meaningful criteria. Gregor and Hevner [13] refined this position, arguing that DSR contributions are strongest when they demonstrate both relevance to a real-world problem and rigour in the design and evaluation process. Sein et al. [14] extended the framework to action design research, encouraging iterative refinement based on real-world feedback.

Table 2.1 below summarises the key works reviewed in this chapter in relation to the study's topic, their areas of focus, the methods used and the gaps from those studies relative to this study.

Table 2.1: Literature Review: Works Discussed in Relation to the Study's Topic

Ref	**Year**	**Study**	**Focus**	**Method**	**Gap Relative to This Study**
[1]	2015	Spreadsheet Error Rates	Formula Error Prevalence	Empirical Survey	No structured database alternative proposed
[2]	2022	Digital Government in Developing Countries	Infrastructure Constraints	Literature Review	No applied prototype or CPI data system
[3]	2019	Data Governance Frameworks	Institutional Data Maturity	Conceptual Review	No applied data system or prototype built
[4]	2019	Defining Digital Transformation	Presentation-First Digitalisation	Qualitative Interview Study	No back-end data system or ETL documented
[7]	2009	Errors in Operational Spreadsheets	Decision-Making Consequences	Empirical	No database solution or CPI application
[9]	2024	Constructing Actionable Insights	Data-to-Insight Gap in BI	Conceptual	No schema, ETL or statistical data prototype
[10]	2012	BI and Analytics	BI Value and Architecture Maturity	Literature Synthesis	No applied prototype, ETL or CPI focus
[11]	2013	Data Warehouse Toolkit	Dimensional Modelling and Star Schema	Technical Reference	No CPI application or developing country focus
[15]	2025	Federated Health Analytics Platform	Layered BI Architecture	Architectural Evaluative	Not inflation or spreadsheet focused; no CPI data
[16]	2023	BI Strategies and Trends	BI Strategy and Evolution	Bibliometric Analysis	No prototype, benchmarking or CPI context
[17]	2023	BI Impact on Analysis Effort	BI Analysis Efficiency	Case Study	Not CPI or statistical system focused
[18]	2024	Public Health Dashboard Principles	Dashboard Design Principles	Scoping Review	No schema, ETL documentation or CPI focus
[19]	2024	Agricultural Trade BI Dashboard	BI Dashboard Application	Applied Design	Limited database and ETL detail; no CPI focus

2.7 Research Gap

The literature reviewed confirms that CPI data is widely used but poorly managed at the infrastructure level, that spreadsheet workflows introduce well-documented risks to accuracy and reproducibility, and that BI tools offer genuine analytical value when supported by sound data architecture. However, no existing study documents a fully transparent, replicable prototype that integrates a structured relational database, a documented ETL pipeline, and interactive BI dashboards specifically for CPI analysis in a developing country institutional context. Works that come closest, such as Adeshina's [15] federated health analytics platform, address BI architecture in public sector settings but do not focus on CPI data, transparent ETL documentation, or the specific institutional constraints of developing country statistical environments. This study addresses that gap directly, producing a documented and replicable prototype that demonstrates how structured data infrastructure can improve the reliability, transparency, and analytical efficiency of CPI monitoring, and that, with appropriate adaptation, offers a transferable model for other major statistical domains.

Chapter 3

Research Design

3.1 Research Methodology

This study adopts a Design Science Research methodology. The choice is deliberate and directly tied to the nature of the problem being addressed. Much of information systems research focuses on understanding and explaining phenomena, documenting why problems exist, analysing their causes, and theorising about their effects. While that kind of inquiry has genuine value, it does not produce a working solution. DSR argues that research can and should build practical artefacts that address real, identified problems, and that the act of building and evaluating those artefacts constitutes a legitimate and rigorous scholarly contribution [12].

This study is fundamentally about solving a problem rather than describing one. The problem, that CPI analysis in developing country institutional contexts relies on error-prone, opaque, and manually intensive workflows, is well established in the literature. What is needed is not further analysis of that problem but a demonstrated alternative. DSR provides the framework for doing exactly that: designing an artefact, implementing it, and evaluating it against the conditions that motivated its creation [13].

The DSR process followed in this study moved through six sequential phases. The first phase involved identifying the problem and gathering requirements from GSS CPI publications and the limitations of existing workflows. The second phase involved designing the artefact through a 3NF relational schema, an ETL architecture, and dashboard wireframes. The third phase was system construction in PostgreSQL and Power BI. The fourth phase involved demonstration and technical testing through SQL queries, execution timings, and dashboard validation. The fifth phase was evaluation through spreadsheet benchmarking, GSS data reconciliation, and scenario-based testing. The sixth and final phase was reflection and communication, documented in Chapters 7 and 8.

3.2 Data Source and Case Context

The primary data source for this study is the Ghana Statistical Service StatsBank platform, through which GSS makes its statistical datasets publicly available for download. Monthly CPI data covering January 2015 to December 2024 was retrieved directly from StatsBank in Excel format. StatsBank provides CPI data in formats other than Excel, but Excel was selected because it reflects the format most commonly encountered by institutional analysts in developing country settings. This kept the workflow aligned with familiar institutional practice. Four separate downloads were made: one for the CPI index values, one for year-on-year inflation rates, one for month-on-month inflation rates, each covering 120 months of national-level data across all product categories, and a fourth providing category-level CPI index values across fifteen COICOP divisions for the same period.

Ghana was selected as the setting for this study for reasons that are both practical and contextually appropriate. The GSS publishes CPI data consistently and makes it freely available, providing a complete dataset without the access barriers that complicate research in other contexts. The ten-year window from 2015 to 2024 captures stable pre-crisis years, a sharp inflationary episode in 2022 and 2023, and early signs of moderation in 2024, making it analytically rich and well-suited to demonstrating the system's capabilities across varied conditions. The system designed in this study is intended to be transferable, and Ghana is the case through which its design, implementation, and performance are demonstrated.

3.3 System Requirements

The functional requirements were derived from the problem statement and objectives set out in Chapter 1. At a minimum, the system needed to store all CPI data from the four StatsBank files in a single structured repository, support querying by period, category, and indicator, retrieve CPI-related values in a consistent structured form, and present results through interactive dashboards without requiring users to manipulate the underlying data manually.

The non-functional requirements reflected the institutional context the system is designed to serve: accuracy reconcilable against official GSS figures, transparency through a fully documented and reproducible pipeline, practicality through freely available tooling deployable without specialist infrastructure, and query performance meaningfully faster than the manual alternatives.

3.4 System Architecture

The system is structured around four layers, each with a distinct role in the data pipeline. The data layer consists of a PostgreSQL v15 relational database containing four tables designed in Third Normal Form: `dim_period`, `dim_category`, `dim_indicator`, and `fact_cpi`. This layer is responsible for structured storage, constraint enforcement, and efficient querying of CPI data.

The integration layer handles data ingestion and transformation. Raw StatsBank Excel files were cleaned and restructured into CSV format, then loaded into a staging table via SQL COPY commands before being inserted into the production schema through JOIN operations that resolved foreign key relationships. This process is fully documented and reproducible through the SQL scripts stored in the project's GitHub repository.

The analytics layer consists of five PostgreSQL views, `vw_headline_cpi`, `vw_category_cpi`, `vw_inflation_rates`, `vw_annual_summary`, and `vw_category_annual_avg`, that pre-compute commonly needed aggregations and serve as the data source for both the Power BI dashboards and the performance benchmark queries. The presentation layer comprises four interactive Power BI dashboards connected directly to the PostgreSQL database, visualising headline CPI trends, category-level indices, year-on-year and month-on-month inflation changes, and multi-period trend exploration with user-controlled filters. Figure 3.1 shows all phases of the architecture from the data layer through to the presentation layer.

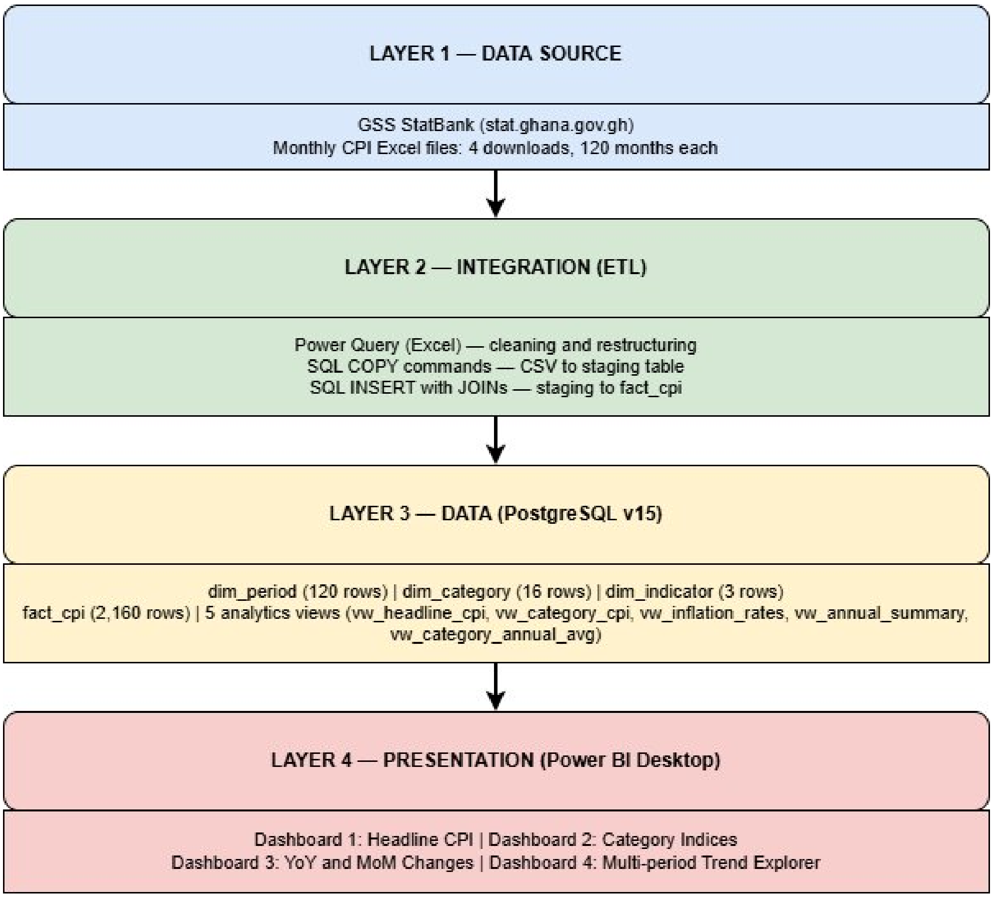

Figure 3.1: System Architecture: CPI Ghana BI Prototype – Four-Layer Design

3.5 Tool Selection

The tools selected for this study were chosen on the basis of accessibility, practicality, and alignment with the resource constraints characteristic of developing country institutional settings. PostgreSQL v15 was selected as the database management system because it is open-source, well-documented, and widely supported, requiring no licensing costs. It provides robust support for relational schema design, constraint enforcement, and time-series querying, all of which are central to this study's requirements.

Microsoft Power BI Desktop was selected for dashboard development because it is freely available, integrates directly with PostgreSQL, and provides a capable visualisation environment without requiring specialist programming expertise. SQL scripts were used for schema creation, data loading, and view definition because they are transparent, version-controllable, and reproducible. Draw.io was used for schema and architecture diagramming because it is free and browser-based. GitHub was used for version control and public code hosting, satisfying the requirement that the system be open to scrutiny and replication.

3.6 Evaluation Strategy

The system was evaluated against the core problems it was designed to address: the unreliability and opacity of spreadsheet-based CPI workflows. Success was defined as the system's ability to deliver accurate, verifiable outputs faster and more reliably than the manual processes it replaces. Four evaluation approaches were used. First, query performance benchmarking measured the execution time of five SQL queries across the analytics layer views. Second, data reconciliation verified the system's outputs against official GSS bulletin figures. Third, a spreadsheet benchmark comparison replicated three analytical tasks manually in Excel and recorded the time taken, providing a direct contrast with the corresponding SQL execution times. Fourth, scenario-based dashboard testing assessed the accuracy and responsiveness of the four Power BI dashboards under realistic analytical conditions.

3.7 Ethical and Practical Considerations

This project uses secondary, publicly available data published by an official national statistics body. The dataset contains no personal or sensitive information and involves no human participants. No formal ethics approval was therefore required, as confirmed by the project ethics screening form submitted alongside the project specification.

Chapter 4

System Implementation

4.1 Artefact Overview

The prototype built in this study comprises five interconnected components: four cleaned and validated source datasets, a PostgreSQL relational database with a 3NF star schema, a documented ETL pipeline, an analytics layer of five SQL views, and four interactive Power BI dashboards. Each component plays a distinct role in the data pipeline, and together they demonstrate how a structured, low-cost system can provide a structured alternative to the manual workflows described in Chapters 1 and 2.

4.2 Data Preparation

The primary data source is the GSS StatsBank platform, from which four datasets were downloaded covering January 2015 to December 2024. The first three downloads contained monthly national-level data for the Consumer Price Index, year-on-year inflation rates, and month-on-month inflation rates respectively, each spanning 120 months. The fourth download provided category-level CPI index values across fifteen COICOP divisions for the same period.

Each file was downloaded in Excel format, as it reflects the format commonly encountered by institutional analysts in this context, and allows the cleaning process to be demonstrated in Power Query, a tool already familiar to the intended user base. Each downloaded file contained metadata rows, regional columns, and junk rows at the bottom that required removal before loading. The first three files were cleaned in Power Query: metadata and junk rows removed, the region column deleted, columns renamed to match the database schema, appropriate data types applied, and two custom columns added to identify the indicator and category for each row. Figure 4.1 shows the cleaninig process of YoY data, with all applied steps shown in the right panel. Similar steps were used in cleaning other data files as well. The cleaned files were exported as CSV for loading into PostgreSQL.

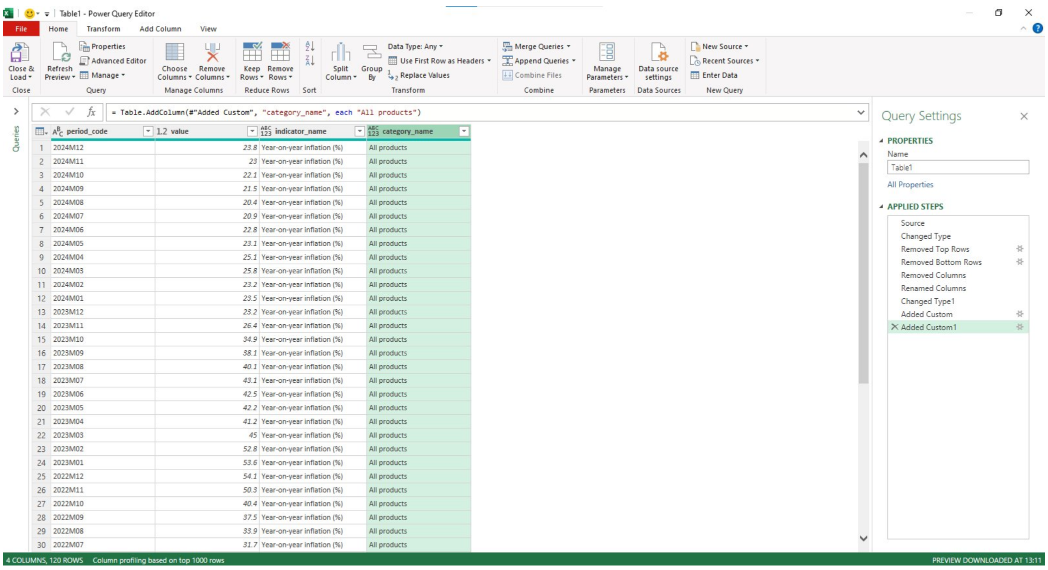

Figure 4.1: Power Query: Cleaned YoY Inflation Data (4 columns, 120 rows, ready for loading)

The fourth file, containing category-level data, was structured differently from the other three. Period codes appeared only on the first row of each monthly group, with subsequent rows leaving that field empty. The file was processed in Power Query to carry forward period codes across all rows within each group, producing a clean three-column CSV containing `period_code`, `category_name`, and `value` across 1,800 rows.

One significant data issue emerged during preparation. Initial cross-referencing of the StatsBank dataset against official GSS monthly bulletins revealed apparent inconsistencies between figures in the two sources. Further investigation established that these inconsistencies were not errors in either source but a consequence of differing base year reference periods across publication formats. GSS bulletins have been published under different base years at different points in time, 2012=100, 2018=100, and 2021=100, meaning that the same underlying price movement is expressed as a different index value depending on which base year anchors the series. StatsBank figures on the other hand, were referenced to the 2021 base year. Once this methodological difference was accounted for, the StatsBank figures and the bulletin figures were found to be fully consistent with each other. This finding underscores the importance of understanding the provenance and methodology of statistical data before drawing conclusions about its accuracy.

4.3 Database Schema Design

The database was implemented in PostgreSQL v15 under the name `cpi_ghana`. The database was organised around a central fact table and three dimension tables, while preserving Third Normal Form in the implemented schema. Third Normal

Form was chosen because it eliminates data redundancy, enforces referential integrity through foreign key constraints, and ensures that each piece of information is stored in exactly one place, qualities that are essential for a system designed to be auditable and reproducible [3].

The three dimension tables each serve a distinct descriptive role. `dim_period` stores the 120 monthly periods from January 2015 to December 2024, generated automatically using PostgreSQL's `generate_series` function to eliminate the risk of manual entry errors. `dim_category` stores the sixteen category values, 'All products' and the fifteen individual COICOP divisions. `dim_indicator` stores the three measurement types: Consumer Price Index, Year-on-year inflation (%), and Month-on-month inflation (%). The central fact table, `fact_cpi`, stores 2,160 rows of measurement data, each referencing a period, a category, and an indicator through foreign keys. Figure 4.2 below is a diagram of the schema, clearly showing the central `fact_cpi`, drawing on data from the other tables through foreign keys. A unique constraint on the combination of `period_id`, `category_id`, and `indicator_id` prevents duplicate entries at the database level, providing a layer of data integrity that spreadsheet workflows cannot replicate.

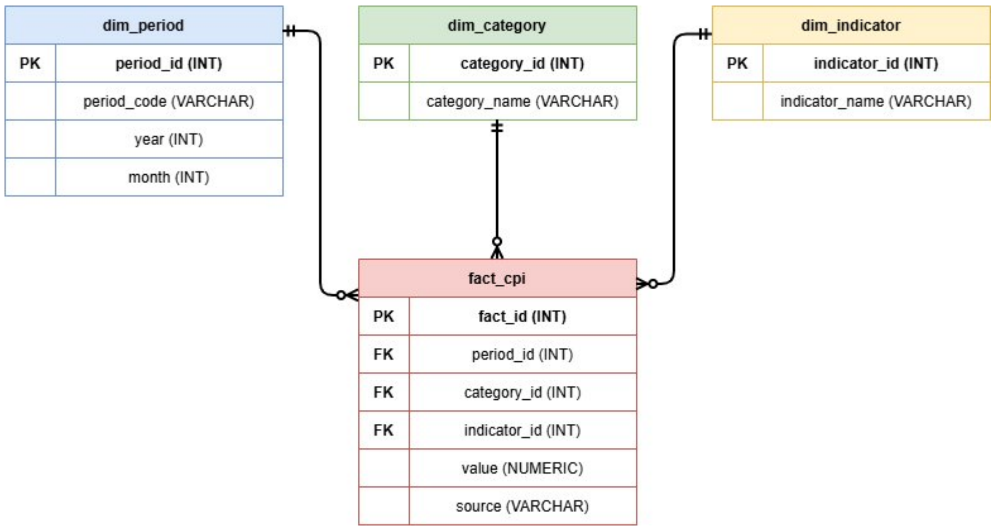

Figure 4.2: `cpi_ghana` Star Schema Design: 3NF with `fact_cpi` and Three Dimension Tables

4.4 ETL Pipeline Implementation

The ETL pipeline was implemented entirely through documented SQL scripts, making every step of the data loading process transparent, version-controlled, and reproducible. The full set of scripts is available in the project GitHub repository (Appendix A). Data loading followed a staging table approach. Rather than loading CSV data directly into `fact_cpi`, a temporary staging table was created to receive the raw CSV data, separating the ingestion step from the transformation step and making errors easier to trace before data is committed to the production schema [3].

The three headline CSV files were loaded into the staging table using PostgreSQL's COPY command, producing 360 rows. The category CSV was handled through a dedicated `staging_category` table, loading 1,800 rows. Data was then transferred into `fact_cpi` using an INSERT statement with three JOIN operations that resolved foreign key IDs from the staging text values. The 360 headline rows were inserted in 110 milliseconds and the 1,800 category rows in 191 milliseconds, producing a verified total of 2,160 rows. The staging tables were then dropped, leaving a clean four-table production schema.

4.5 Analytics Layer

Five SQL views were created to form the analytics layer between the raw database and the Power BI dashboards. Views were chosen over direct table queries because they pre-compute commonly needed aggregations, present clean and meaningful field names to Power BI, and can be updated centrally without requiring changes to dashboard configuration. `vw_headline_cpi` returns the full 120-month CPI index series for all products. `vw_category_cpi` returns the same series broken down by individual category. `vw_inflation_rates` presents year-on-year and month-on-month rates side by side for each period using conditional aggregation. `vw_annual_summary` calculates yearly average CPI, YoY, and MoM figures for the full decade. `vw_category_annual_avg` calculates yearly average CPI per category, supporting category-level trend analysis across years. All five views were created in a single script in pgAdmin, which executed in 91 milliseconds, as shown in Figure 4.3.

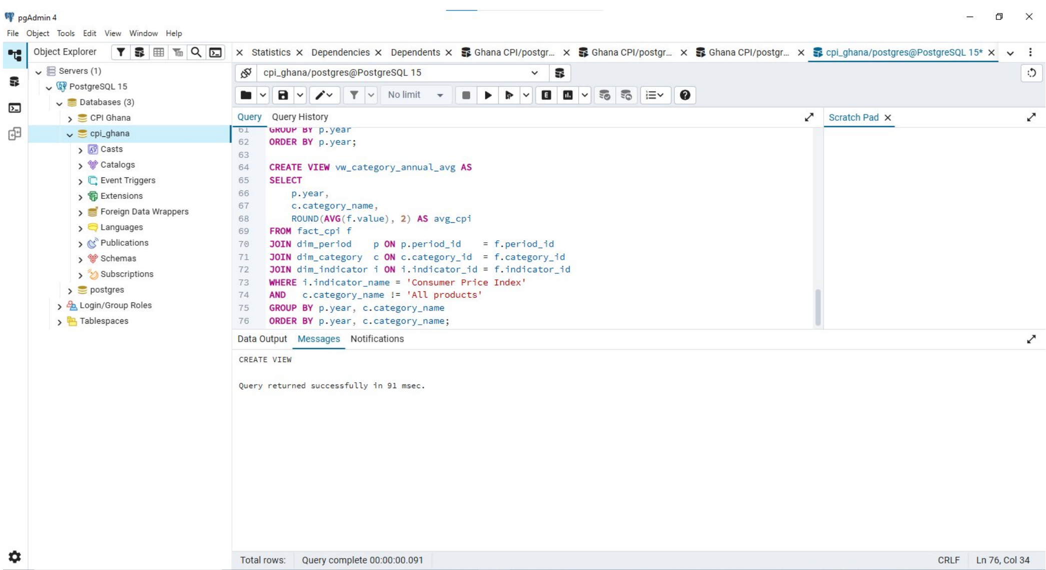

Figure 4.3: Analytics Layer: Five SQL Views Created in pgAdmin (91ms execution)

4.6 Power BI Integration

Power BI Desktop was connected to the `cpi_ghana` PostgreSQL database using the built-in PostgreSQL connector. Figure 4.4 shows that upon loading the four production tables, Power BI automatically detected the foreign key relationships defined in the schema and rendered them as a star schema in the Model view, with `fact_cpi` at the centre and the three dimension tables surrounding it. This auto-detection confirmed that the schema was correctly structured and that the relationships were unambiguous. All four dashboards draw their data through this live connection, meaning any update to the database is reflected in the dashboards upon refresh, a significant advantage over static spreadsheet-based reporting.

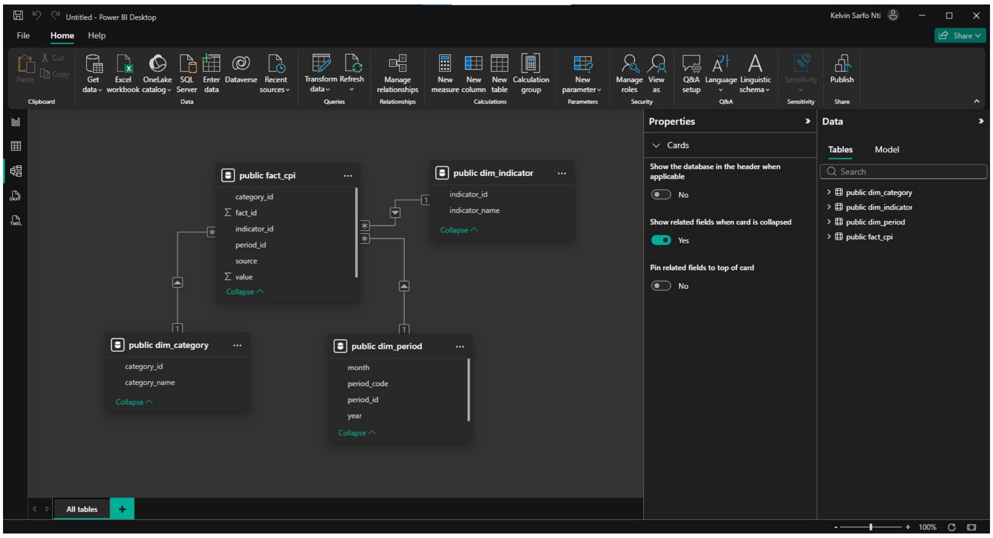

Figure 4.4: Power BI Model View: Star Schema Auto-detected from PostgreSQL Schema

4.7 Dashboard Development

Four interactive dashboards were developed in Power BI Desktop, each targeting a specific analytical use case. Dashboard 1 presents the headline CPI index trend from 2015 to 2024 as a line chart, filtered to the Consumer Price Index indicator and the All products category. From Figure 4.5, the chart clearly shows the stable growth period from 2015 to 2021 and the sharp acceleration in 2022 and 2023.

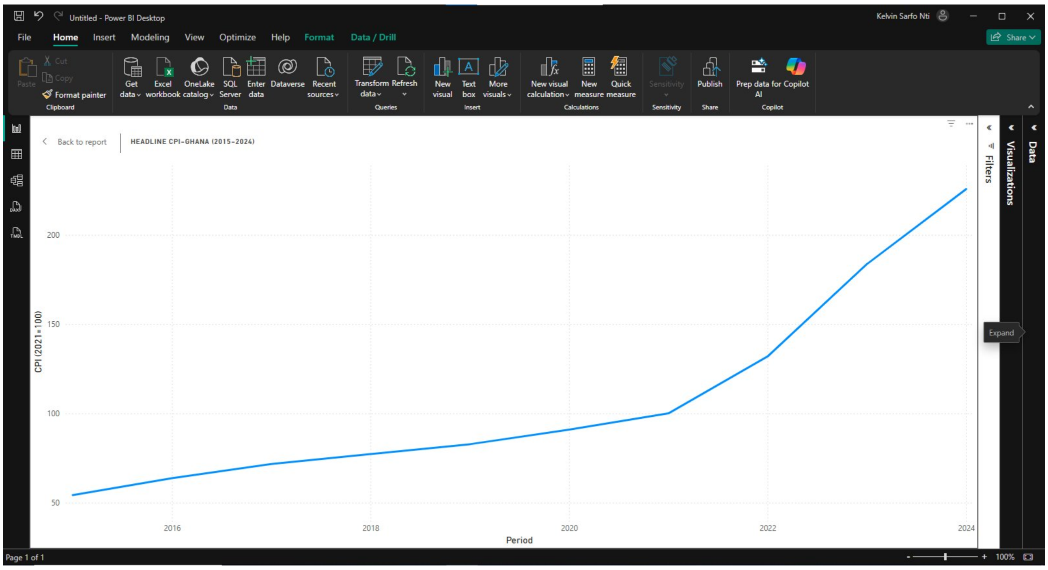

Figure 4.5: Dashboard 1: Headline CPI Trend, Ghana (2015–2024)

Dashboard 2 presents category-level CPI indices across the same period, with thirteen individual COICOP categories shown as separate lines, shown in Figure 4.6. Composite categories such as All products, Food and non-alcoholic beverages, and Non-food were excluded to avoid duplication and keep the comparison focused on distinct divisions. Dashboard 3 presents year-on-year and month-on-month inflation rates on the same chart, distinguished by colour, making it immediately visible that YoY rates spiked dramatically in 2022 and 2023 while MoM rates remained comparatively stable throughout, evident from Figure 4.7.

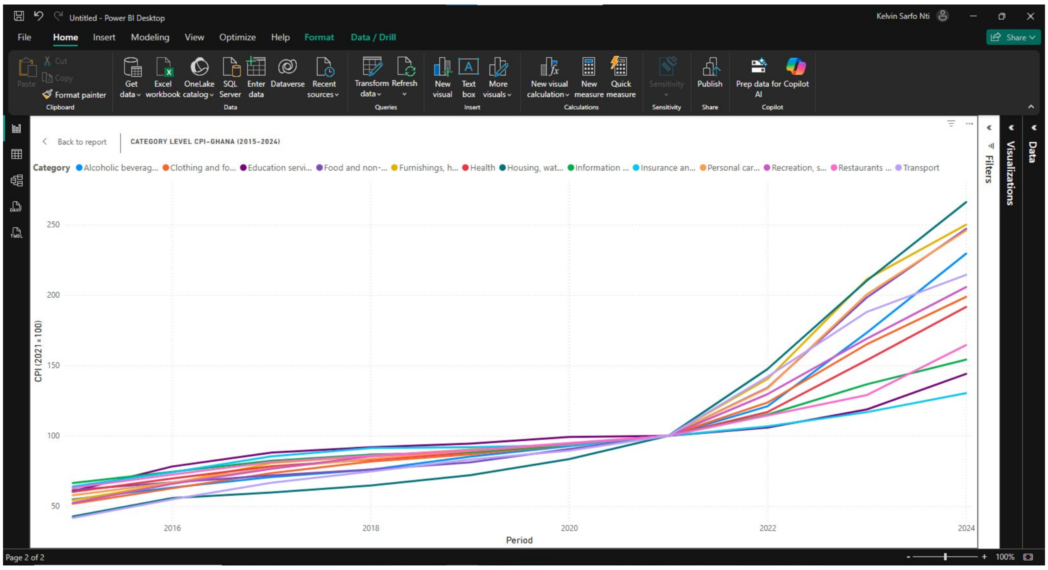

Figure 4.6: Dashboard 2: Category-Level CPI Indices, Ghana (2015–2024)

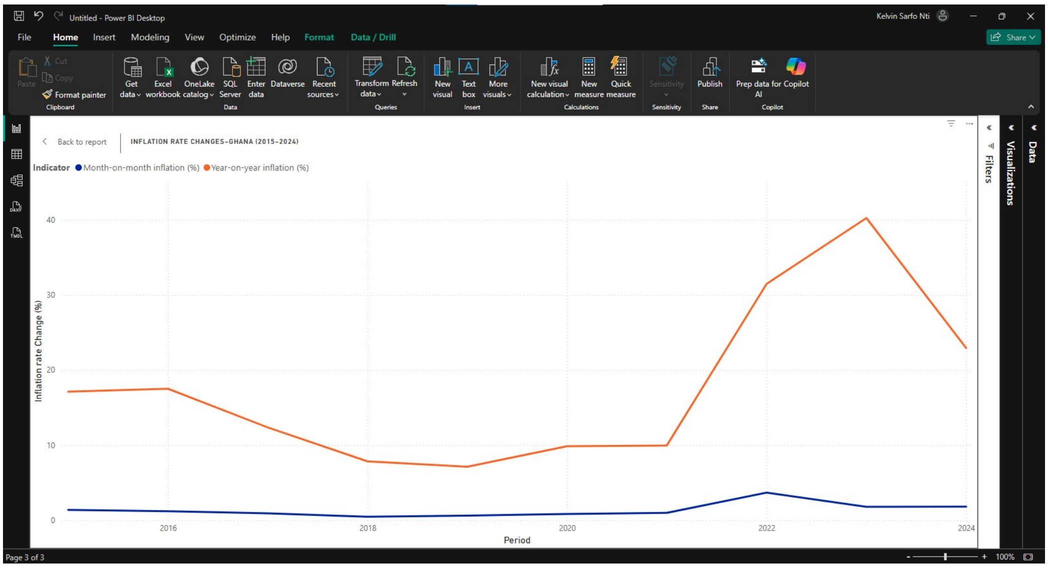

Figure 4.7: Dashboard 3: YoY and MoM Inflation Rates, Ghana (2015–2024)

Dashboard 4 is the most interactive of the four. It presents category-level CPI trends with two user-controlled slicers, one for period range and one for category selection, allowing users to isolate specific periods or categories for focused analysis.Figure 4.8 shows the movement of Food and Transport across two years in a chart. This dashboard directly addresses the comparison limitations of bulletin-based analysis described in Chapter 1, where cross-category or cross-period comparisons required opening multiple documents simultaneously.

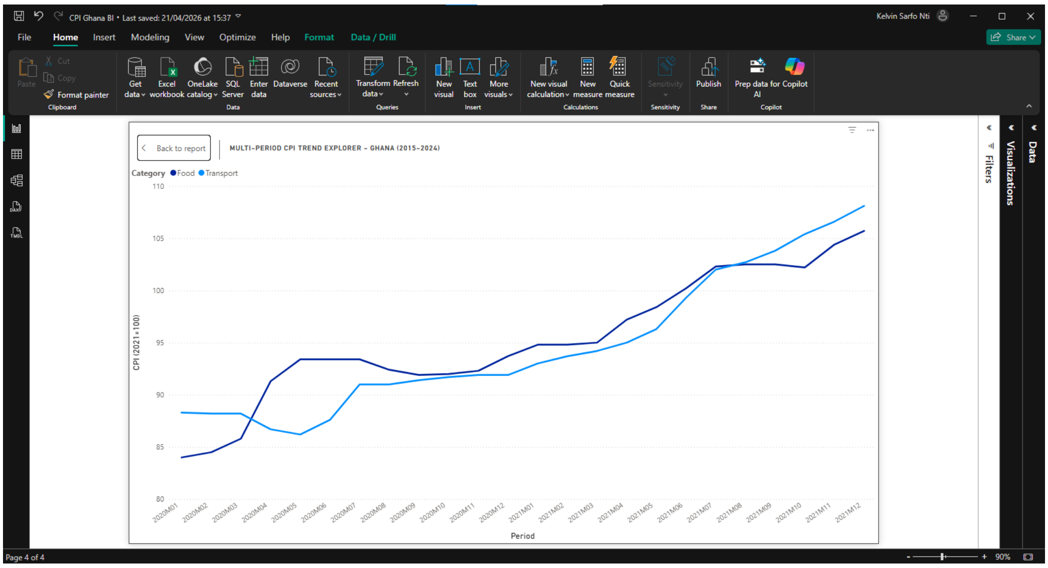

Figure 4.8: Dashboard 4: Filters set to Food and Transport for 2020 and 2021

Chapter 5

Testing and Results

5.1 Chapter Introduction

This chapter covers data integrity verification, query performance benchmarking, spreadsheet comparison, and scenario-based dashboard testing, followed by the key analytical findings from the CPI dataset.

5.2 Data Integrity Results

Figure 5.1 shows the first validation step, confirmed the total row count in `fact_cpi` using a COUNT query, which returned 2,160 rows in 130 milliseconds. A more detailed verification query joined `fact_cpi` to the dimension tables and grouped results by indicator and category, confirming eighteen distinct combinations each with exactly 120 rows, consistent with 120 months of data across three indicators and sixteen categories. A sample of figures from the database was cross-referenced against official GSS monthly bulletins for three separate years, confirming full consistency with the published source data after accounting for the base year difference documented in Chapter 4. No constraint violations, duplicate rows, or missing values were identified at any stage of loading or verification.

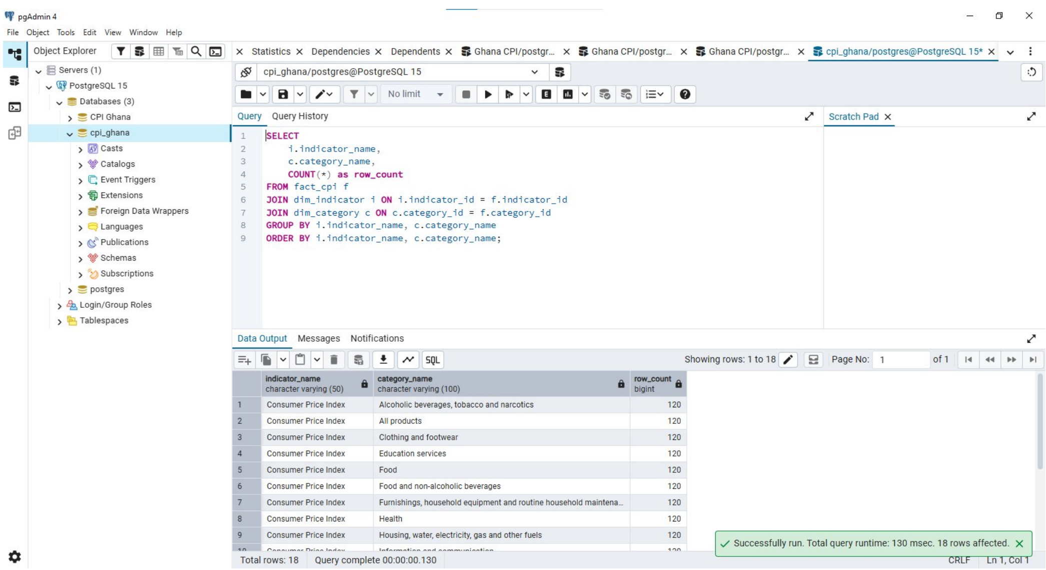

Figure 5.1: Data Integrity Verification: 18 Indicator/Category Combinations, Each with 120 Rows

5.3 Query Performance Results

Table 5.1 presents the results of five benchmark queries, which were run against the analytics layer views to measure execution performance, showing the various queries, the number of rows of data returned and the execution time.

Table 5.1: SQL Query Performance Benchmark Results

Query	Description	Rows	Execution Time
Q1	Headline CPI full series	120	573ms
Q2	Annual inflation summary	10	265ms
Q3	Peak YoY by year	10	187ms
Q4	Highest category average CPI 2023	15	148ms
Q5	Full category trend	1,800	294ms

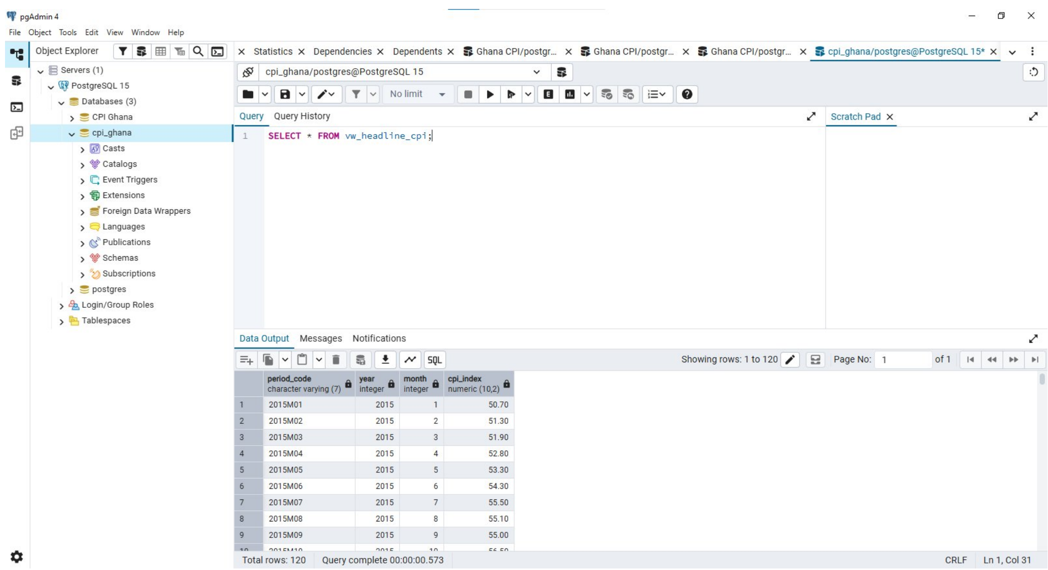

Figure 5.2: Q1 Benchmark: `vw_headline_cpi` Query Result (120 rows, 573ms)

All five queries completed in under 600 milliseconds, with the largest result set of 1,800 rows returned in 294 milliseconds. Figure 5.2 shows the results of query 1, which returned the CPI values for all 120 months in the decade span used in this study, and executed in 573 milliseconds. These results demonstrate that the analytics layer delivers responsive performance across a range of result set sizes and query complexities. The results are consistent with the expectations of a locally deployed prototype of this scale, and confirm that the view-based analytics layer serves as an effective intermediary between the raw schema and the dashboards.

5.4 Spreadsheet Comparison Results

To provide a direct contrast with the SQL results, Table 5.2 shows the three equivalent analytical tasks that were replicated manually in Microsoft Excel using the raw StatsBank files, as well as the time taken for each task to be completed. The results were then compared against the corresponding SQL execution time.

Table 5.2: Spreadsheet versus SQL Benchmark Comparison

Task	Excel Time	SQL Time
Annual summary (all 10 years)	4 mins 23 secs	0.157s
Transport category yearly averages	2 mins 3 secs	0.141s
Peak YoY by year	1 min 2 secs	0.120s

The results confirm a substantial performance advantage for the SQL-based approach across all three tasks. The annual summary task, which required opening three separate files, calculating yearly averages for each indicator, and combining them into a single table, took over four minutes manually compared to 0.157 seconds via SQL. Beyond speed, the SQL results are produced from a documented, version-controlled script that can be executed at any time to produce an identical output. The manual Excel process introduces the risk of formula errors and version inconsistencies at every step [1], and provides no equivalent guarantee of reproducibility.

5.5 Dashboard Scenario Testing Results

Three scenario-based tests were conducted. Scenario 1 confirmed that Dashboard 1's headline CPI line chart displayed a consistent upward trend from 2015 to 2024, with values matching the verified database and consistent with GSS published figures. Figure 5.3 shows the result of Scenario 2, which filtered Transport and 2022 to 2023 into Dashboard 4, and the dashboard updated within seconds, accurately reflecting Transport's position as one of the highest-inflation categories during that period. Scenario 3 confirmed that Dashboard 3's YoY rate line peaked visibly in 2022 and 2023 while the MoM line remained comparatively flat, consistent with the Q3 benchmark result. All three scenarios met expectations.

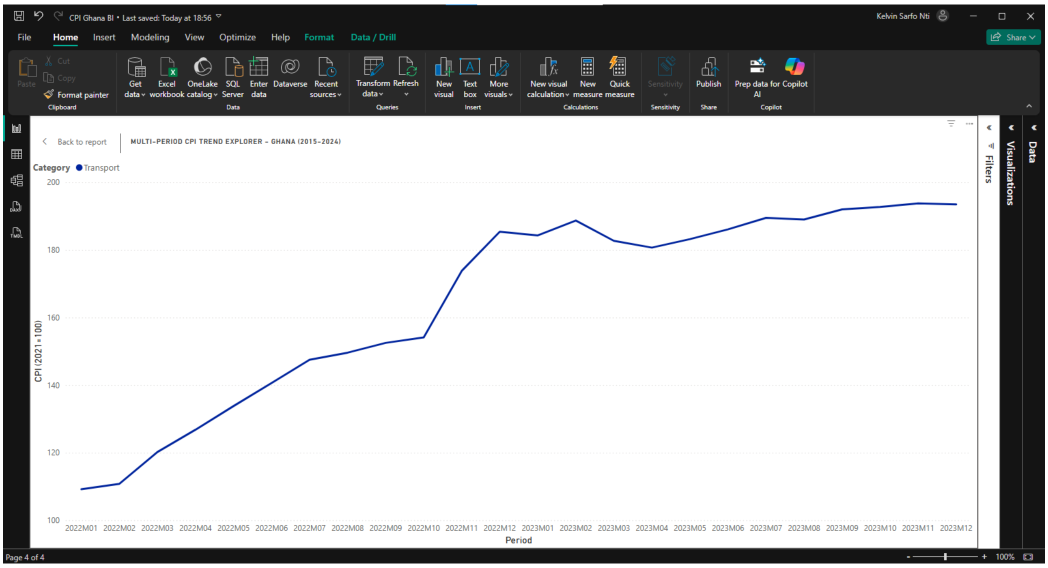

Figure 5.3: Scenario 2 Result: Transport Category Filter Applied (2022–2023)

5.6 Headline CPI Trends

The headline CPI data covering January 2015 to December 2024 reveals a clear and telling inflation narrative. From 2015 through to 2020, Ghana's Consumer Price Index rose steadily and at a relatively controlled pace, with the index moving from an average of 54.27 in 2015 to 100.00 in 2021, the base year, representing gradual but manageable price growth over six years. The picture changed sharply from 2022 onwards. The annual average CPI rose to 131.93 in 2022 and accelerated further to 183.68 in 2023, the highest on record in the dataset. This rapid escalation is consistent with the severe inflationary pressures Ghana experienced during this period. Early signs of moderation are visible in the 2024 data, where the year-on-year rate began to ease, suggesting the initial effects of stabilisation measures were beginning to filter through to consumer prices.

5.7 Category-Level Dynamics

The category-level analysis reveals that the inflationary pressures of 2022 to 2023 were felt across virtually all product divisions, though with notable variation in intensity. Transport recorded the highest average CPI index value in 2023 at 188.02, reflecting its direct exposure to fuel price increases and exchange rate movements. Alcoholic beverages, tobacco, and narcotics followed at 173.34, with Recreation, sport, and culture and Clothing and footwear also recording elevated indices above 165. At the other end of the spectrum, Education services recorded the lowest average CPI in 2023 at 118.70, followed by Insurance and financial services at 116.83. The relative stability of Education is particularly notable when compared with more market-sensitive categories like Transport or Food. This suggests that categories respond differently to broader price pressures, illustrating a key finding. Broad headline inflation figures can obscure significant variation at the category level, and a system capable of disaggregating inflation by category provides meaningfully richer analytical value than headline figures alone.

5.8 Inflation Rate Patterns

The year-on-year and month-on-month rates, presented together in Dashboard 3, reveal the speed and intensity of Ghana's inflationary episode with particular clarity. The YoY rate remained relatively contained under 20% for most of the 2015 to 2021 period, but rose dramatically from 2022 onwards. The Q3 benchmark query confirmed a peak single-month YoY rate of 54.10% in 2022, the highest on record in the dataset, followed by 53.60% in 2023 before beginning a visible moderation to 25.80% in 2024. The month-on-month rate tells a complementary story: across the full decade, MoM rates remained comparatively low and stable, rarely exceeding 3% in any single month even during the crisis period. This contrast reflects the cumulative nature of the inflationary episode. Prices rose persistently month after month, and it is that persistence rather than any single extreme monthly jump that produced the record annual figures.

5.9 Summary of Findings

The testing results confirm that the prototype performs reliably and accurately across all evaluation criteria. The data integrity checks produced no violations, the query performance results fall well within acceptable response thresholds, and the scenario-based dashboard tests all passed. The spreadsheet comparison placed these results in direct contrast, as tasks requiring between one and four minutes of manual effort in Excel were completed by the SQL system in under 200 milliseconds each. The analytical findings in turn demonstrate that the system does what it was designed to do, supporting the identification of meaningful insights about Ghana's inflation trajectory at both the headline and category level, in a format that is faster, more transparent, and more reproducible than the manual workflows.

Chapter 6

Discussion

The problem identified in Chapter 1 had two dimensions: spreadsheet dependency and architectural opacity. The results in Chapter 5 suggest the prototype addresses both, though with important qualifications. On the first dimension, the benchmark comparison confirms that structured SQL queries dramatically outperform manual Excel workflows on the tasks tested. More significantly, every result produced by the system is traceable to a version-controlled script and verifiable against the original source data. That combination of speed, transparency, and reproducibility is an advantage the manual approach cannot provide consistently [3].

On the second dimension, the system's documentation, published through GitHub alongside the full SQL codebase, provides the kind of architectural transparency that platforms like the IMF DataMapper do not. Anyone wishing to understand how a particular output was produced can follow the pipeline from the raw StatsBank CSV through the staging table, the production schema, the analytics views, and into the dashboard. That is a meaningful departure from the opacity that characterises most public inflation dashboards.

In addition, the results affirm several claims from the literature. The spreadsheet comparison supports Panko's [1] and Powell et al.'s [7] arguments about the practical risks of manual workflows, that the time cost is real, and the reproducibility cost is greater still. The constraint-enforced schema supports Abraham et al.'s [3] position that structured data governance is foundational to institutional data maturity. The auto-detected star schema in Power BI confirms Kimball and Ross's [11] argument that dimensional modelling enables efficient and consistent reporting. The system's performance is consistent with Petersen et al.'s [17] finding that BI deployments produce measurable reductions in analytical effort even in resource-constrained settings.

Furthermore, the strongest elements of the build are its reproducibility and its transparency. The entire pipeline, from data download through to dashboard output, can be reconstructed from the SQL scripts and Power Query steps documented in the GitHub repository. The star schema design proved structurally sound, as no constraint violations occurred during loading, and Power BI detected the relationships correctly without manual configuration. The analytics views serve their purpose effectively, pre-computing aggregations that would otherwise require repeated ad-hoc queries and providing a clean interface between the database and the dashboards. The category-level analysis, particularly the divergence between Transport and Education in 2023, demonstrates clear analytical value that would be difficult to surface quickly through bulletin-based manual analysis.

The prototype's architecture is not Ghana-specific. The schema design, the ETL approach, and the dashboard framework are built around a general pattern of regularly published time-series statistical datasets with national and category-level dimensions. With adaptation of the data ingestion layer and the category structure, the same system could be applied to CPI or equivalent datasets in any developing country context with consistent national statistics publications. This transferability was a deliberate design priority, informed by Nielsen and Sahay [2] and Mergel et al. [4], who establish that the institutional constraints driving spreadsheet dependency are not unique to Ghana. At the same time, successful transfer would still depend on local data quality, publication consistency, and organisational capacity.

Chapter 7

Conclusion

7.1 Summary of What Was Built

This study set out to design, implement, and evaluate a prototype relational database and business intelligence system for analysing Consumer Price Inflation data in Ghana. The study has met that aim at the level of a working and evaluated prototype.

The prototype comprises a 3NF star schema implemented in PostgreSQL v15, a documented ETL pipeline built through staged SQL scripts, an analytics layer of five SQL views, and four interactive Power BI dashboards. Every component is publicly available through the project's GitHub repository, and the full pipeline from raw StatsBank download to dashboard output is reproducible by other users and institutions with access to similar tools and source data.

7.2 Main Findings

The central finding is that a structured, transparent, and low-cost alternative to spreadsheet-based CPI analysis is both technically feasible and practically achievable in a developing country institutional context. The benchmark results confirm that tasks requiring between one and four minutes of manual effort in spreadsheet workflows are completed by the SQL analytics layer in under 200 milliseconds. Every output is traceable back to its source data and produced from a version-controlled script, making it less exposed to formula errors and version inconsistencies that characterise manual analysis. The category-level results further demonstrate that the system surfaces insights, such as the sharp divergence between Transport and Education inflation in 2023, that headline figures alone would obscure.

7.3 Limitations

The prototype has genuine limitations that must be acknowledged honestly. The data preparation process remains partially manual, as files are downloaded from StatsBank and cleaned in Power Query before loading into PostgreSQL. While this process is documented and reproducible, manual steps retain a degree of human error risk that a fully automated ingestion pipeline would eliminate. The analysis is limited to national-level data, with regional breakdowns excluded from the current implementation, which constrains the depth of sub-national analysis the system can support. The dataset covers a single decade, which, while analytically rich, limits the system's capacity to reveal longer-term structural inflation patterns. Finally, the evaluation relied on scenario-based testing rather than formal user testing with institutional stakeholders, which provides a relatively limited assessment of real-world usability and workflow fit.

7.4 Future Directions

Several directions present themselves for future development. The most immediate priority would be automating the data ingestion pipeline by replacing manual download and cleaning steps with a scheduled process that retrieves updated figures directly from the GSS StatsBank platform. Incorporating regional CPI data would substantially increase analytical depth. Extending the dataset beyond the current decade would strengthen the system's value for long-term trend analysis. A more rigorous evaluation framework, involving formal usability testing with analysts at institutions such as the GSS or the Ministry of Finance, would provide evidence of practical effectiveness that scenario-based testing alone cannot deliver.

7.5 Final Concluding Statement

The research gap identified in Chapter 2, that no existing study documents a fully transparent, replicable prototype integrating a relational database, ETL pipeline, and BI dashboards specifically for CPI analysis in a developing country context, has been addressed through the design and evaluation of the presented prototype. The prototype built in this study demonstrates that structured data infrastructure can meaningfully improve the reliability, transparency, and analytical efficiency of inflation monitoring in resource-constrained settings, and that the same design principles are transferable to other statistical domains and institutional environments.

Chapter 8

Critical Self-Evaluation

8.1 What Went Well

Across the full arc of this project, a significant amount went right. The final database schema proved structurally sound, as the 3NF star schema arrangement was appropriate for the project's requirements, and the foreign key relationships held up without issue throughout the entire data loading and querying process. The ETL pipeline executed cleanly, the analytics views returned accurate results, and all four dashboards connected to and reflected the underlying data correctly. The benchmark results confirmed that the system performed well within acceptable response times across all query types.

Beyond the technical outcomes, the project benefitted from strong version control discipline. SQL scripts, cleaned data files, diagrams, and the Power BI file were committed to GitHub systematically, producing a transparent and auditable project record that directly supports the reproducibility claims made throughout the dissertation. Perhaps the most telling indicator of progress is the contrast between the first prototype build and the rebuild. The first build was completed under time pressure and with limited familiarity with the tools. The rebuild, undertaken with a clearer plan and accumulated experience, was smoother, faster, and more methodical. That progression from uncertainty to competence is itself evidence of genuine learning.

8.2 Challenges Faced

The most significant challenge was the base year discrepancy identified during data preparation. What initially appeared to be inconsistencies between the StatsBank data and the official GSS bulletins required careful investigation before the methodological explanation became clear. Resolving this correctly rather than dismissing the apparent inconsistency was important both for data integrity and for the intellectual honesty of the study. The initial rebuild decision was also a difficult one to make: recognising mid-project that the first build was not good enough and committing to starting again required a frank assessment of the work produced so far. The diagrams were also completed later in the project than they should have been, which meant some architectural decisions were documented retrospectively rather than informing the build in real time.

8.3 What Would Be Done Differently

The most significant change would be to invest more time at the start in becoming more familiar with each tool before attempting to build anything. Learning PostgreSQL, Power Query, Power BI, and GitHub in depth at the same time created friction that slowed early progress and contributed to the need for a full rebuild. While version control was maintained throughout the project, establishing a more structured commit discipline from the very first day would have produced an even richer project history, with finer documentation of each design decision as it was made.

8.4 Skills Developed

This project developed practical competence across several technical and academic domains. On the technical side, the most significant growth was in SQL view design, star schema modelling, Power Query transformation, and Power BI data modelling. On the academic side, it strengthened structured argument-making, evidence-based evaluation, and the discipline of keeping a dissertation aligned to what was actually built. Applying DSR as a methodology and being held to its phased structure also developed a more systematic approach to managing complex technical work.

8.5 Final Reflection

The most transferable lesson from this project is a simple one: more appropriate tools often exist beyond the familiar ones initially chosen, provided the effort is made to find and learn them. The instinct at the start of the project was to reach for the familiar, the spreadsheet. Working through the process of designing a relational schema, building an ETL pipeline, and connecting it to a BI platform demonstrated concretely that the effort of learning a better solution pays off in ways that compound. The system that resulted is faster, more reliable, and more transparent than a spreadsheet workflow would typically be for the same dataset. DSR proved to be the right methodological framework for this kind of work: its insistence that research should produce solutions, not just descriptions of problems, shaped both the project design and the way its outcomes are understood and reported.

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Appendix A

GitHub Repository

The complete project codebase, including all SQL scripts, cleaned data files, Draw.io diagrams and the Power BI file, is publicly available at the following repository:

Repository: `https://github.com/kntisarfo/Final-Dissertation-Project`

The repository is organised into the following folders:

sql/ – All eleven numbered SQL scripts in execution order.

Data/ – Cleaned CSV files exported from Power Query, used as ETL inputs.

diagrams/ – Draw.io schema diagram and four-layer architecture diagram.

dashboard/ – Power BI Desktop file (`CPI_Ghana_BI.pbix`).

Appendix B

Data Source

All CPI data used in this study was sourced from the Ghana Statistical Service (GSS) StatsBank platform, the official open-access statistical repository of the GSS. The datasets are publicly available for download in multiple formats including Excel, CSV and JSON.

Ghana Statistical Service StatsBank: `https://statsbank.statsghana.gov.gh`

The following four datasets were downloaded for this study, covering January 2015 to December 2024:

Consumer Price Index (CPI) – national monthly CPI index values (120 rows, base year 2021=100)

Year-on-Year Inflation Rate (%) – national monthly YoY figures (120 rows)

Month-on-Month Inflation Rate (%) – national monthly MoM figures (120 rows)

Category-Level CPI – fifteen COICOP division indices across all 120 months (1,800 rows)

All datasets are referenced to the 2021=100 base year. The base year difference relative to earlier GSS bulletin formats (2012=100 and 2018=100) is discussed in Chapter 4, Section 4.2.

Appendix C

Project Specification

This appendix contains the approved project specification submitted to the University of the West of Scotland in March 2026, as signed off by supervisor Dr. Nurudeen Oladehinbo Salau. It covers the project title, abstract, justification and aim, objectives, literature review, methodology and work plan.

Project Title

Design, Implementation and Evaluation of a Relational Database and Business Intelligence Prototype for Consumer Price Inflation Analysis in Ghana.

Abstract

Consumer price inflation is a critical macroeconomic indicator used by governments, central banks, businesses, and households to guide economic decision-making. In Ghana, the Ghana Statistical Service (GSS) publishes monthly Consumer Price Index (CPI) statistics, that provide national and category level inflation measures. While these datasets are publicly available, institutional inflation monitoring workflows in many developing countries environments remain heavily dependent on spreadsheet-based tools. Such approaches are widely recognised as vulnerable to formula errors, weak version control, limited auditability and poor scalability when handling large and complex time series data.

Although public inflation dashboards and open-data portals exist internationally, such as the International Monetary Fund's (IMF) DataMapper and the World Bank's Macro Poverty Outlook, these platforms typically prioritise end-user visualisation over transparent documentation of the underlying information-system architectures that support data ingestion, validation, storage, and reproducible analysis. Moreover, digital governance research further suggests that developing countries public institutions often face structural constraints in establishing formalised data infrastructures capable of supporting long term analytical consistency. Thus, a few to none of these platforms explicitly show how inflation data is processed, managed, and transformed for policy use.

Hence, this project proposes the design, implementation, and evaluation of a prototype relational database and business intelligence (BI) dashboard system for analysing CPI data in Ghana. Monthly CPI publications from the Ghana Statistical Service will be ingested into a structured relational schema to enable efficient querying, validation, and exploratory analysis. A set of interactive dashboards will be developed to visualise headline inflation trends, category-level indices, and temporal comparisons across key inflation periods.

The project uses a design science research method to help with requirements analysis, system building, and technical evaluation. Testing will include performance benchmarks that can be repeated, validation of the audit trail, and workflow simulations based on real-life scenarios. The methodology is based on established DSR frameworks for the creation and evaluation of IT artifacts in practical, applied contexts. The goal is to create a well-documented and reproducible prototype system that comes with useful, transferable ideas for improving the flow of inflation data in environments with limited technical resources.

Justification and Overall Aim

Consumer Price Inflation directly influences central bank policy decisions, wage negotiations, investment strategy, and social welfare planning. In Ghana, GSS produces monthly CPI releases that inform a wide range of economic stakeholders. These datasets offer rich potential for inflation trend analysis, yet the systems used to process and interpret these data often fall short of institutional needs. In many developing countries public sector institutions, CPI analysis continues to be carried out using spreadsheet-based tools. While spreadsheets are familiar and accessible, they are widely recognised as limited in handling large or evolving time series datasets. Version control problems, inconsistent formula application, lack of auditability, and scalability issues reduce confidence in the accuracy, reproducibility, and efficiency of spreadsheet-based workflows [1].

Moreover, these workflows often lack formal data validation processes, thus limiting error detection and correction to informal and non-standardised practices.

Beyond spreadsheet risk, broader digital transformation literature highlights systemic challenges in building sustainable public sector data infrastructures in developing countries contexts [2]. Data governance research emphasises that structured relational databases and formalised data lifecycle management are fundamental to institutional data maturity [3]. However, in many cases, dashboard adoption precedes back-end architectural reform, resulting in presentation focused digitalisation without transparent schema governance [4].

Although global organisations such as the IMF and World Bank publish inflation dashboards and CPI trend visualisations, these platforms focus primarily on dissemination, not on transparent documentation of the system architecture behind the analysis. They rarely describe how CPI data is extracted, validated, transformed, stored, and used in decision-making systems. This reflects a broader gap in academic and technical literature particularly in the developing country context, where few studies document the full lifecycle of CPI data within institutional workflows, especially through database backed and business intelligence (BI) enabled systems.

Therefore, this project proposal addresses that gap by proposing the design, implementation, and evaluation of a relational database and BI dashboard prototype for CPI analysis in developing country contexts, like Ghana. It will ingest publicly available CPI datasets from the GSS and convert them into a structured, queryable format. Interactive dashboards will provide users with visual access to inflation trends, year-on-year (YoY) and month-on-month (MoM) changes, and category level dynamics. By applying a design science research (DSR) approach, this study aims to demonstrate whether lightweight but structured system interventions, databases and BI dashboards in this scenario, can improve data quality, analytical efficiency, and reporting reproducibility for inflation monitoring in resource constrained environments.

Overall Aim: To design, implement, and evaluate a prototype relational database and business intelligence system for analysing Consumer Price Inflation data in Ghana.

Objectives

The project will pursue the following structured and measurable objectives, aligned with a design-build-evaluate methodology:

- Review and analyse the structure, format, and consistency of monthly CPI publications released by the GSS between 2015 and 2024.
- Specify system requirements, both functional and non-functional, for a CPI analysis platform suited to institutional users.
- Design a relational database schema using Third Normal Form (3NF) to structure CPI data for efficient storage, validation, and querying.
- Implement the schema in PostgreSQL v15, configuring tables, primary and foreign keys, constraints, and indexes optimised for time-series analysis.
- Ingest CPI datasets using structured ETL workflows, combining SQL scripts and Power BI Desktop dataflows.
- Develop four interactive dashboards in Power BI to visualise: headline CPI across time, category-level inflation indices, year-on-year and month-on-month changes, and multi-period trend exploration with filterable parameters.

- Benchmark performance by comparing dashboard report generation time, user interaction speed, and data validation effort against spreadsheet-based workflows.
- Evaluate system quality through query performance tests, reconciliation with official GSS figures, and structured scenario-based testing to assess reproducibility and usability.
- Reflect critically on design and implementation decisions, noting system limitations, technical trade-offs, and recommendations for institutional deployment or future feature expansion.

Work Plan

The project was planned over a 16-week period structured around the six DSR phases.

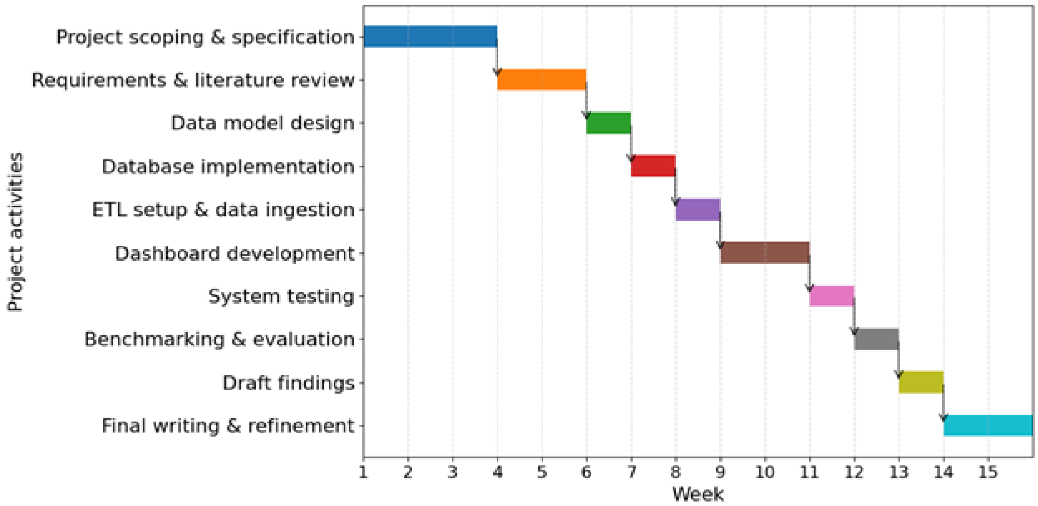

Figure C.1: Project Work Plan - Gantt Chart (15-Week DSR Timeline)